Chapter 13
Sustainable Variables Implemented in the Protection of Ecological Flow as a Strategy to Mitigate the Effects of Climate Change

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Abstract Ecological flow, defined as the water flow regime necessary to sustain ecosystem health, is a critical factor in global water resource management. However, human activities and the lack of institutional and community involvement have significantly impacted its stability. This study conducts a systematic review of 126 articles from the SciELO and Scopus databases, categorizing them based on thematic similarities, with a focus on ecological flow impacts, aquatic biota, and strategies for improvement. The research methodology utilizes the Parsifal tool, enabling hypothesis development and leading to three key findings: (1) identification of the primary indicators influencing ecological flow and aquatic biota, (2) comparative analysis of vulnerability levels among different water bodies, and (3) recognition of the deficit in the integrated application of sustainable conservation mechanisms. The results highlight those variations in alteration parameters, affected water bodies, and the effectiveness of sustainable strategies depend on the specific environmental conditions of each study area. Aligned with Sustainable Development Goal (SDG) 13: Climate Action, the study underscores the importance of ecological flow management not only for aquatic ecosystem conservation but also for climate change mitigation and adaptation. The protection and restoration of water bodies should be prioritized as fundamental strategies to counteract climate change effects, integrating environmental conservation with community well-being. These findings emphasize the urgent need for policies and actions that align with global sustainability objectives, particularly in the context of climate resilience.

Keywords Ecological flow · Aquatic biota · Sustainability · Water body

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1 Introduction

Water is a fundamental resource for both humans and ecosystems, playing a key role in biotic and abiotic development worldwide (Westwood et al., 2017). Based on its physical, chemical, and biological composition, various ecological, climatic, hydraulic, morphodynamic, and other variables are attributed, which describe the behavior of the flow in the presence of different alterations that may lead to serious consequences in the future (Qin et al., 2016). This problem arises from the impact of industrial activities, agricultural expansion, solid waste pollution, and discharges containing high levels of toxic compounds (Gleeson & Richter, 2018). The significance of this problem is due to the intervention of productive activities from industry, agricultural growth, pollution from solid waste, and discharges with high loads of toxic compounds, which daily reflect an increase in the loss of natural ecosystems worldwide.

It is crucial to acknowledge that the protection and sustainable management of water resources are integral to combating climate change and its impacts. The adverse effects of industrial activities, agricultural expansion, and pollution on water ecosystems exacerbate the challenges posed by climate change, leading to more severe and frequent alterations in water flow and quality. Addressing these issues requires concerted efforts to implement sustainable practices, reduce pollution, and enhance the resilience of water ecosystems. By aligning with the objectives of SDG 13, we can work towards mitigating climate change, protecting vital water resources, and ensuring the health and sustainability of ecosystems globally.

Ecological flow is one of the most impacted factors, as the diversity of aquatic species and ecosystem development are constrained by variations in physical, chemical, and biological water parameters. Sustainable ecological flow management is key to mitigating climate change impacts. Changes in physical, chemical, and biological parameters affect biodiversity, alter aquatic ecosystems, and intensify climate-related challenges. Implementing sustainable practices and reducing pollution are vital to improving the resilience of aquatic ecosystems and ensuring their ability to adapt to changing climate conditions. These actions help mitigate the effects of climate change and protect water resources and biodiversity in an integrated manner (Tibúrcio et al., 2016). These parameters daily yield alarming figures regarding changes in the life cycle of aquatic organisms, the decline of their populations in various bodies of water, and the extinction or morphological modification of terrestrial and aquatic habitats (Wang et al., 2018). Likewise, it is taken into account that the variability in ecological flow affects the usual structure of ecosystems, their composition, and their distribution over time (Chen et al., 2019). This leads to simultaneous water discharges that impact the development and survival of species in various aquatic ecosystems, where there is high exposure to characteristics that contribute to water stability across all bodies of water, thereby limiting the normal course of ecological flow under different ecosystem scenarios (Tsheboeng, 2020).

In various regions such as Asia, North America, and Latin America, the lack of effective water management strategies, hydrological data, and adequate monitoring tools has hindered ecological flow conservation (Boavida et al., 2020). Likewise, there is a reflected lack of support from government entities and other institutions responsible for promoting and conserving water biodiversity in various regions (Gleeson & Richter, 2018). In this same context, the use and exploitation of water by communities are considered, where the lack of social strategies for proper water management and aquatic biota conservation is evident, aiming to reduce the scarcity of this resource in various cities and aquatic ecosystems (Magalhães Junior & Curiel, 2017).

Several studies have explored sustainable mechanisms for conserving ecological flow. Some of the evaluation mechanisms for flow promote research tools, regulations, monitoring, and formal education, which encompass economic, social, and environmental components, both in areas of productive development and in emerging risk zones (Liu et al., 2011; Yan et al., 2018). However, the application of methodologies has been limited due to insufficient data collection and the difficulty of accessing studies in various databases. Therefore, there is a need to apply procedures that synthesize existing information about the current state of ecological flow in different regions, as this is an essential factor for aquatic biodiversity indices, biological interactions, and environmental processes in various natural regions (Garcia et al., 2020).

In response to this issue, this study is proposed as a valid global source of information, aiming to conduct a systematic literature review. This review compiles research focused on the intervention of external factors influencing flow and the biotic environment in general. The strategy applied in this review was formulated through three key questions: (Q1) What physical, chemical, and biological variables indicate alterations in ecological flow? (Q2) In what types of water bodies is there a greater impact on aquatic species due to variations in ecological flow? (Q3) What sustainable variables or methods are implemented to establish rehabilitation mechanisms for the current state of ecological flow and aquatic biota? These questions determined pertinent information within each article regarding the main causes and critical areas of water impact, allowing for an analysis of the study's object from scientific, environmental, and social perspectives to identify limitations within the research framework applied to the conservation of ecological flow and aquatic ecosystems.

2 Methodology

This study aims to determine the main causes and critical areas of impact on ecological flow to identify sustainable variables implemented for the protection and conservation of aquatic biota. To achieve this objective, it is essential to conduct a systematic review of information to select articles focused on addressing these issues. This review facilitated data collection through a systematic selection process that is shown in Fig. 13.1.

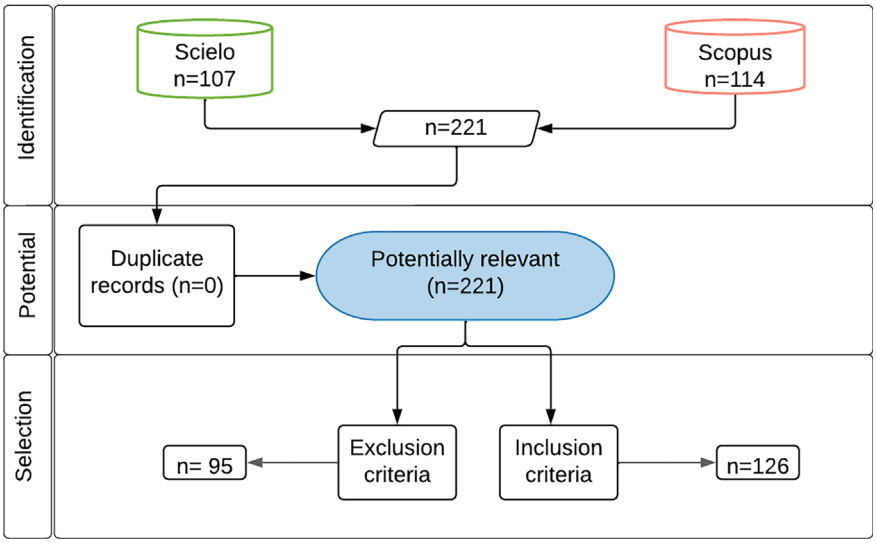

Fig. 13.1 Methodological development of the systematic review

2.1 Acquisition of Articles

Scielo and Scopus were selected for their extensive global research coverage, offering broader access to relevant studies compared to other databases. Scopus primarily collects works in English, while Scielo focuses on Spanish publications (Thürer et al., 2018). To obtain a representative number of articles and high-quality information, search criteria were established, highlighting English-language articles with the following synonyms in their title, abstract, or keywords: "Impact on aquatic biota," "Impact on ecological flow," "Physical water parameters," "Biological water parameters," "Chemical water parameters," "Conservation of aquatic biota," "Ecological flow," and "Sustainability." Additionally, a restriction was set to include only articles published from the year 2000 to the present, resulting in a total of 221 articles.

After this selection, 69 articles from the Scielo database and 26 articles from Scopus were discarded, resulting in a final selection of 126 articles, composed of 38 articles from the Scielo database and 88 articles from the Scopus database.

3 Results

The results obtained from the systematic review address the previously posed questions Q1, Q2, and Q3. Each section will provide an in-depth analysis of the information contributed by the selected articles and will also present a compilation of statistics on the research distribution around each hypothesis through comparative

tables. This study not only addresses the issue of ecological flow and its impact on aquatic biodiversity but also contributes directly to global efforts to combat climate change, in line with SDG 13, Climate Action. By identifying and promoting sustainable variables and conservation methods, this work supports the mitigation of climate change effects and the adaptation of aquatic ecosystems to new climatic conditions. Thus, the research not only adds scientific value but also reinforces the importance of sustainable actions to protect our water resources and biodiversity, aligning with sustainable development goals and the fight against climate change. Over a period of 21 years, the selected articles demonstrated a growing trend, showing progress in research on ecological flow and its influence on aquatic biota. It was found that 38% of the articles were published between 2018 and 2020. The total distribution of research papers by Geographic Location was North America 19%, South America 15.9%, Europe 14.3%, Asia 35.7%, and Oceanic Islands 2.4%.

3.1 Literature Analysis

Physical, chemical, and biological variables are crucial in evaluating the water quality of ecological flow. Implementing sustainable strategies for the protection and conservation of ecological flow helps mitigate the effects of climate change, promoting the resilience of aquatic ecosystems to changing climatic conditions. In this way, conservation efforts are aligned with SDG 13, ensuring that the actions taken not only protect water resources but also contribute to sustainable development and environmental protection on a global scale (García De Jalón, 2008).

As shown in Table 13.1, temperature was the most studied variable compared to others, accounting for 59.5% of the articles. Conversely, solids were the least investigated variable in this review, representing only 11.9% of the articles related to physical parameters.

Table 13.2 shows that dissolved oxygen was the most studied variable compared to the others, representing 29.4% of the articles. Similarly, COD and BOD were the least researched variables in this review, representing 11.6% of the articles related to chemical parameters.

Table 13.3 shows that fish were the most studied variable compared to others, representing 48% of the articles.

Table 13.1 Physical parameters reported in the papers

Physical Variables	Number of Articles	Percentage (%)
Temperature	25	59.5
Salinity	6	14.2
Turbidity	6	14.2
Solids (TS, TSS, SS)	5	11.9

Table 13.2 Chemical parameters addressed in the papers

Chemical Variables	Number of Articles	Percentage (%)
Dissolved oxygen	10	29.4
pH	6	17.6
Nitrites and nitrates	4	11.7
Phosphates and phosphorus	4	11.7
Metals or metalloids	3	8.8
Iones	3	8.8
COD	2	5.8
BOD	2	5.8

Table 13.3 Biological parameters addressed in the papers

Biological Variables	Number of Articles	Percentage (%
Fish	25	48
Macroinvertebrates	12	23
Plants	12	23
Pathogenic microorganisms	2	3.8
Mammals, reptiles, and birds	1	2.2

Table 13.4 Other causes of ecological flow alteration addressed in the papers

Other impact variables	Number of articles	Percentage (%)
Climatic alterations	43	35.2
Megastructures	43	35.2
Productive activities	23	18.8
Climate change	7	5.7
Geomorphological causes	6	4.9

Table 13.4 shows that climate alterations were the most studied variable compared to the others, representing 35.2% of the articles. Similarly, geomorphological causes were the least researched variables in this review, representing 4.9% of the articles related to other alteration factors.

3.2 Impact Due to Variations in Ecological Flow

Water bodies can be analyzed at different spatial and temporal scales. Understanding their ecological flow dynamics allows for the early detection of environmental disturbances (Pantoja, 2017). Consequently, analyzing this problem question provides

the theoretical foundation for determining which mechanisms for the protection or conservation of water and ecological resources should be applied to aquatic ecosystems.

Several studies have examined different water bodies, with a particular focus on rivers. A total of 54 studies highlight that water flow is a key factor influencing aquatic species' behavior and habitat condition (Mantel et al., 2010). The research conducted by Theodoropoulos et al. (2019) demonstrated that optimal living conditions were a result of the hydraulic characteristics of the riparian bed. For instance, the limnophilic cockroach was found in deep sections where the flow maintained low velocities, and the Evrotas fish preferred moderately deep habitats with low velocities. However, unlike these species, the small cacho fish adapted to shallow depths where the flow had high velocities (Theodoropoulos et al., 2019). The impact of flow variability and habitat conditions is also reflected during different stages of organism growth. Thus, bioindicators can serve as indicators of the most affected habitats in rivers, providing a measure of ecological health in response to disturbances (Boavida et al., 2020; Carolli et al., 2021).

Similarly, streams or brooks are considered another type of water body. Compared to rivers, they differ in flow quantity and the diversity of aquatic organisms, predominantly inhabited by macroinvertebrates, macrophytes, and small fish (Rosa et al., 2015). Thus, 16 studies confirmed the importance of streams as areas capable of providing protection, refuge, and food, essential for maintaining the biotic and abiotic conditions of various habitats (Tibúrcio et al., 2016). On the other hand, Patterson et al. (2017) conducted an eco-deficit monitoring of stream habitats, focusing on riffle, riffle-run, and pool formations. The riffle and riffle-run groups exhibited a small eco-deficit (2.2%), indicating a low level of anthropogenic impact. In contrast, pools had a higher eco-deficit (4.5%), associated with the presence of urban areas and agricultural activities. This led to a decline in stream quality, significantly impacting fish and macroinvertebrate communities and contributing to stress and mortality among aquatic species (Patterson et al., 2017).

Wetlands, like rivers, are water bodies that support diverse habitats and are key indicators of biodiversity and ecosystem services. However, they face significant impacts from socioeconomic developments. In this context, five studies assessed the impact of unsustainable activities in various wetlands worldwide, showing that the most significant impacts are caused by agricultural, mining, tourism, and urban practices. For example, the wetland known as "El Pantanal" in Brazil has reduced its area by 17% due to the expansion of agricultural frontiers, resulting in invaluable wildlife losses and an annual environmental degradation rate of 2.3% (Alho, 2008). Likewise, Tan et al. (2012) report that in the Zoige Wetland in China, soil degradation was observed due to a 62.52% increase in intensive grazing activities (Tan et al., 2012). These values signify vulnerability and stress on the ecological flow, delaying the optimal development of habitats and ecosystems within wetlands (Talukdar et al., 2020).

Finally, estuaries are one of the water bodies most affected by climatic variability, as increased temperature, precipitation, and productive activities lead to land misuse, pollution, alterations in salinity levels, deforestation, and reduction in ecological flow.

Similarly, Martínez et al. (2011) confirmed through their study of the estuaries on the coast of Mexico that various fish groups have restricted their distribution in the water column due to intensive fishing and depletion of dissolved oxygen. This highlights the inadequate environmental control in these water bodies and suggests scenarios with significant coastal habitat loss (Martínez et al., 2011).

Currently, natural and anthropogenic alterations affecting ecological flow have reflected multiple impacts on aquatic communities. However, various studies have been required to determine mechanisms or tools for conserving flow and aquatic biota, so that water resources can restore their natural conditions and be ecologically healthy.

Twenty-five articles establish a direct link between conducting research and providing a clearer understanding of aquatic habitat components (Liu et al., 2011). The data collected from these research efforts ensure a comprehensive approach to understanding aquatic biodiversity and the ecological flow necessary for its sustainability (Butakka et al., 2014; Katopodis, 2005). Additionally, environmental modeling and simulation play a crucial role in analyzing ecological dynamics, predominantly through habitat and water balance models. These models use ecological data to replicate ideal conditions for ecological flow, supporting the well-being of aquatic organisms (Qin et al., 2016). Restoring, whether partially or fully, the characteristics of an ecosystem or its ecological flow is vital to maintaining the integrity and growth of the aquatic community (da Costa et al., 2015). Eleven studies highlight the importance of developing management alternatives focused on habitat recovery and resilience measures. These strategies, combined with other methodologies, contribute to sustaining biodiversity and maintaining acceptable water quality for aquatic species. The variability of ecological flow is often unpredictable and challenging to understand. Consequently, ten studies have examined the implementation of monitoring techniques to create temporal and comparative scenarios that describe typical flow conditions (Castro et al., 2013; Pedone-Bonfim et al., 2018). Including biota in all water body monitoring plans is crucial for comprehensive assessments that inform decision-making about their condition (Weerts & Cyrus, 2008). Such processes should engage governmental agencies, research centers, and various professionals to strengthen conservation efforts and the management of aquatic habitats holistically (Martínez et al., 2011).

Identifying protected ecological and water zones is essential for enhancing ecological processes. Eight studies confirm that maintaining aquatic species within buffer zones is a fundamental approach (Haas et al., 2014; Roets et al., 2008). This practice supports ecological connectivity, safeguards native species, and prevents negative impacts on water characteristics, nutrient flow, and pollutant removal, ensuring a sustainable and healthy future for aquatic ecosystems (Beaton & Bradford, 2013; McKenna, 2019). Integrating ecological parameters with policies that support water conservation and protection is emphasized by two studies, which stress the need for sustainable practices that balance economic development and ecological flow to preserve aquatic life (Bradford, 2008). Current regulations in some countries or ecosystems may rely on outdated and ineffective methodologies. Therefore, it is

recommended to incorporate scientific research in the review and development of new regulations to strengthen watershed management (da Costa et al., 2015).

Among the different strategies for protecting ecological flow, research stands out as the most widely used method, representing 47.8% of the reviewed studies. In contrast, monitoring remains the least explored approach, with only 7.6% representation. Other significant methods include regulation (12.8%), civil structures (11.9%), hydrological information (11.1%), and environmental indicators (8.5%), indicating a varied but uneven focus on the conservation of aquatic biota.

4 Discussion

In the first place, according to Yan et al. (2018), physical parameters such as temperature, SST, salinity, conductivity, and turbidity are used to establish water quality concepts in various water bodies. In this context, temperature stands out as a variable influencing both biotic and abiotic development in ecological flow. However, Araújo et al. (2013) found that changes in some physical variables, especially temperature, do not have significant repercussions on the biotic level, as living conditions remain stable for most aquatic organisms studied. An example challenging this assertion is found in the Qingshui River in China, where species like C. idella, Xenocypris argentea, Hypophthalmichthys nobilis, and H. molitrix, which typically spawn at average temperatures of 18–23 °C, might experience disruptions in their proliferation processes due to temperature changes (Li et al., 2019; Popa et al., 2020). Similarly, Da Rosa & Garcia (2015) linked climatic seasons with temperature changes in different areas, observing a 56% increase and a 60% decrease in epiphyte populations during the winter season, leading to a thermal stability range from −8 °C to 14 °C. However, to counter this study, Wang et al. (2020) demonstrated that temperature variation does not directly influence the development of organisms in water sources, as this change remains an essential factor for maintaining aquatic ecosystems (Wang et al., 2020).

Relating chemical parameters such as dissolved oxygen, pH, nitrites, nitrates, phosphates, COD, BOD, ions, and metals in ecological flow, a connection is identified with environmental impacts caused by anthropogenic activities, which pose a risk to the balance of aquatic ecosystems (Upadhyay et al., 2011). Among the mentioned parameters, several authors agree that dissolved oxygen is the only factor that significantly influences the chemical characteristics of water. This assertion is supported by a study conducted in the El Miño and Lima estuaries in Portugal, where sampling of dissolved oxygen at the river's entrance and near the mouth to the sea recorded values of 9.5 mg O_2/L and 7.5 mg O_2/L respectively. This variation was confirmed through the correlation between this variable and the presence of BOD, COD, and heavy metals (Oliveira et al., 2019). However, Walsh & Wepener (2009) contradict this information with another variable, pH, which involves a comprehensive analysis of chemical elements negatively influencing ecological flow. This theory is supported by research in the Crocodile and Magalies Rivers in South Africa, where pH levels

were directly related to high concentrations of inorganic nitrogen and COD present in different river sections (Walsh & Wepener, 2009).

Biological variables present in all water bodies are influenced by both physical and chemical factors. This is reflected in the previously mentioned cases of the Qingshui River in China and the El Miño and Lima estuaries in Portugal, where a decrease in aquatic organism population was observed due to variations in temperature and dissolved oxygen resulting from human activities. To counter this, Guerra et al. (2016) in their study of the San Juan River in Mexico identified variable temperatures, reduced discharges, and suspended solids affecting 12 fish species. They confirmed the equitable influence between physical and chemical parameters on the presence/absence of aquatic species and the execution of their biological functions in different river beds (Guerra et al., 2016).

When discussing other alterations that significantly influence ecological flow conditions, reservoirs and climate changes stand out as factors that affect flow both directly and indirectly. In this context, Ma et al. (2018) identified, in their study of 62 dam projects in China, a detrimental effect on river biodiversity due to species extinction and the lack of regulation and monitoring of ecological flow in vulnerable areas (Ma et al., 2019). However, other studies, such as Kuriqi et al. (2020), which propose the construction of dams to enhance flow regulation, have highlighted successful outcomes due to the effective application of water management processes, avoiding disruptions in the usual development of aquatic species (Kuriqi et al., 2020).

Additionally, climate alterations result in variations in temperature and precipitation, indicating varying degrees of impact on flow and aquatic communities (Xu et al., 2014). Brown et al. (2018) recorded an increase in benthic organisms and certain fish species as river flow increased, while some local tree species showed improved quality of life due to nutrient accumulation in the soil resulting from decreased flow in certain sections (Brown et al., 2018). Therefore, it is evident that climatic variation in some regions stands out as a factor that does not consistently have positive or negative effects on the biotic development of different aquatic ecosystems.

It is evident that various water bodies are being affected by the expansion of industrial areas. Among the most significantly impacted factors is the ecological flow, which plays a crucial role in the growth of endemic ecosystems and the proliferation of aquatic species. Rivers, in particular, are among the most vulnerable areas. According to Lee et al. (2013), changes in the presence of species such as C. splendidus and P. nigra in the Geum River in Korea were observed due to variations in flow velocity, bed depth, and substrate size. These changes led to hydrodynamic and morphodynamic alterations in habitat stability over extended periods (Lee et al., 2013). On the other hand, Pal et al. (2020) conducted a study on the Taihu Lake wetland in China, where 36 species disappeared over 57 years due to human activities. Their findings corroborated that, over shorter periods, significant changes in the formation and ecological stability of these water bodies can occur (Pal & Sarda, 2021). This underscores the impact of ecosystem loss in relation to the duration of disturbance. However, Han et al. (2012) found through their study that organisms in streams are not affected by water variability, and thus, there is no significant habitat loss related to the duration of different seasons (Han et al., 2012).

The implementation of sustainable mechanisms for the conservation of ecological flow and aquatic biota, as discussed in the reviewed articles, reveals the potential for deploying highly effective tools for managing water flow in response to social and environmental needs. However, Mwedzi et al. (2016) emphasize a significant issue: the application of these conservation mechanisms often remains isolated rather than integrated. This fragmentation results in a lack of cohesive strategies for managing both ecological flow and aquatic biota, which undermines overall effectiveness. Moreover, economic constraints are a major barrier to the successful implementation of these environmental strategies. Perni et al. (2012) investigated public interest in wetland restoration projects in Sydney and found that the community's willingness to invest was approximately €57.37 per person per year. This figure reflects the financial limitations faced by local populations and governments in high-vulnerability areas. Such economic challenges, coupled with insufficient integration of conservation tools, highlight the need for more coordinated and well-funded approaches. Leite et al. (2016) further support this view by pointing out that the success of conservation initiatives is often dependent on social and governmental willingness to invest in vulnerable regions. Effective management of ecological flow and aquatic biota requires not only the application of individual conservation tools but also the integration of these tools into a comprehensive strategy. This strategy should address both the economic and practical aspects of conservation to ensure sustainable outcomes.

5 Implementation Tips

Protecting ecological flow is essential for conserving aquatic ecosystems and mitigating the effects of climate change. Effective implementation requires adopting strategies that integrate monitoring, regulation, restoration, sustainable water management, technological advancements, and community participation. First, it is crucial to establish continuous assessment and monitoring systems for ecological flow, considering physical, chemical, and biological water variables. The use of hydrological and ecohydraulic models will help anticipate changes in water flow and their impacts on biodiversity. Additionally, implementing environmental indicators will facilitate the identification of alterations in aquatic ecosystems, supporting decision-making based on scientific evidence. From a regulatory perspective, strengthening legal frameworks for ecological flow management is necessary, ensuring that policies are formulated based on scientific criteria and aligned with SDG 13 (Climate Action). The participation of governmental institutions and local communities in the development and enforcement of these regulations will ensure an integrated water resource management approach. This will help balance water use for productive activities with the need to conserve ecosystems. Restoration and conservation of aquatic ecosystems play a key role in protecting ecological flow. Reforestation of watersheds improves water retention and reduces erosion, stabilizing water bodies' flow. Similarly, restoring buffer zones and implementing green

infrastructure, such as artificial wetlands, will enhance water quality and increase ecosystem resilience to climate change. Another fundamental aspect is sustainable water management, promoting responsible use in agricultural, industrial, and urban sectors. Encouraging the adoption of water treatment and reuse technologies, along with incentives for practices that reduce water waste, is essential. Education and awareness campaigns on the importance of ecological flow should be prioritized, fostering knowledge about its impact on biodiversity and community well-being. Technological advancements also play a crucial role in conserving ecological flow. The application of environmental simulation tools and hydrological modeling will help design more precise water management strategies. Additionally, technologies such as remote sensing and real-time monitoring sensors will facilitate the evaluation of changes in water quantity and quality, enabling a rapid response to environmental threats. Finally, community participation is essential to ensuring the success of any ecological flow conservation strategy. Developing awareness campaigns and environmental education programs will encourage communities to engage in sustainable water management. Involving local residents in monitoring and conservation projects will strengthen their commitment to protecting water resources, ensuring a more effective and equitable approach to ecological flow management.

6 Conclusions

Physical and chemical parameters are essential for evaluating changes in ecological flow, providing thresholds that reflect the impact of natural and anthropogenic processes on water quality and flow variability. These parameters include metrics such as temperature, dissolved oxygen, pH, and concentrations of various ions and pollutants. While these indicators are useful, they only offer a partial view of the ecological health of water bodies.

Biological parameters, however, offer a more comprehensive and precise assessment of ecological flow alterations. They integrate physical and chemical factors and focus on the health, diversity, and abundance of aquatic organisms. This holistic approach enables a more accurate characterization and simulation of ecological conditions, which is crucial for effective management and conservation.

Water bodies worldwide are increasingly subjected to human and industrial pressures, leading to significant environmental impacts. Rivers and wetlands are particularly vulnerable, as they are often exploited to meet the needs of growing populations and economic activities. This exploitation results in a range of issues, including habitat degradation, loss of biodiversity, and reduced ecological flow. The consequences are evident in the decline of species populations and the alteration of natural processes that sustain aquatic ecosystems.

Sustainable mechanisms for rehabilitating ecological flow and aquatic biota include a range of strategies: research, regulatory frameworks, monitoring programs, protected areas, and environmental simulation. Each mechanism contributes to the overall goal of conserving and restoring ecological health. Research provides the

foundational knowledge needed to understand ecological processes and impacts. Regulatory frameworks establish legal standards and guidelines for managing water resources. Monitoring programs track changes in water quality and biota, while protected areas offer safe havens for species and habitats. Environmental simulation models help predict the effects of various interventions and guide decision-making.

Although these mechanisms are effective, their implementation often lacks integration. Instead of a cohesive strategy, they are frequently applied in isolation, reducing their overall impact. This fragmentation limits their ability to address the complex challenges faced by aquatic ecosystems comprehensively. To overcome these limitations, there is a pressing need for new, integrated management strategies that combine these mechanisms into a cohesive framework. Such approaches should consider the interconnectedness of physical, chemical, and biological factors and their combined impact on ecological health.

The success of conservation and rehabilitation strategies depends on the unique characteristics of each water body. Geographic and environmental variations affect ecological flow, requiring customized approaches and region-specific data for effective management. Adaptation and customization of conservation practices are essential for addressing the unique challenges of each water body.

Finally, the successful implementation of these strategies requires strong support from government agencies, research institutions, and local communities. Effective water management and regulation depend on collaborative efforts that balance economic, social, and environmental considerations. By fostering partnerships and engaging stakeholders, it is possible to achieve sustainable management of water resources, prevent further biodiversity loss, and ensure the long-term health of aquatic ecosystems. This collective effort is crucial for preserving the ecological integrity of our water bodies and securing a sustainable future for all aquatic life.

References

Alho, C. J. R. (2008). Biodiversity of the Pantanal: Response to seasonal flooding regime and to environmental degradation. *Brazilian Journal of Biology, 68*(4 SUPPL), 957–966. https://doi.org/10.1590/S1519-69842008000500005

Araújo, F. G., Santos, A. B. I., & Albieri, R. J. (2013). Avaliando a similaridade das assembléias de peixes acima e abaixo de uma barragem de um reservatório Neotropical com obliteração parcial. *Brazilian Journal of Biology, 73*(4), 727–736. https://doi.org/10.1590/S1519-69842013000400007

Beaton, A. D., & Bradford, A. (2013). Demonstration of a methodology for setting ecological flow and water level targets. *Canadian Water Resources Journal, 38*(4), 296–310. https://doi.org/10.1080/07011784.2013.830371

Boavida, I., Caetano, L., & Pinheiro, A. N. (2020). E-flows to reduce the hydropeaking impacts on the Iberian barbel (Luciobarbus bocagei) habitat. An effectiveness assessment based on the COSH Tool application. *Science of the Total Environment, 699*, 134209. https://doi.org/10.1016/j.scitotenv.2019.134209

Bradford, A. (2008). An ecological flow assessment framework: Building a bridge to implementation in Canada. *Canadian Water Resources Journal, 33*(3), 215–232. https://doi.org/10.4296/cwrj3303215

Brown, C., Joubert, A., Tlou, T., Birkhead, A., Marneweck, G., Paxton, B., & Singh, A. (2018). The pongola floodplain, South Africa - part 2: Holistic environmental flows assessment. *Water SA, 44*(4), 746–759. https://doi.org/10.4314/wsa.v44i4.22

Butakka, C. M. d. M., Gomes, L. C., & Takeda, A. M. (2014). Estrutura taxonômica e numérica de Chironomidae (Diptera) em diferentes habitats de uma várzea neotropical. *Iheringia - Serie Zoologia, 104*(3), 314–322. https://doi.org/10.1590/1678-476620141043314322

Carolli, M., Gelmini, F., Pellegrini, S., Deriu, M., & Zolezzi, G. (2021). Prioritizing reaches for restoration in a regulated Alpine river: Locally driven versus hydro-morphologically based actions. *River Research and Applications, 37*(1), 17–32. https://doi.org/10.1002/rra.3737

Castro, D. M. P., Hughes, R. M., & Callisto, M. (2013). Influence of peak flow changes on the macroinvertebrate drift downstream of a Brazilian hydroelectric dam. *Brazilian Journal of Biology, 73*(4), 775–782. https://doi.org/10.1590/S1519-69842013000400013

Chen, A., et al. (2019). Bridging gaps between environmental flows theory and practices in China. *Water Science and Engineering, 12*(4), 284–292. https://doi.org/10.1016/j.wse.2019.12.002

da Costa, M. R., Mattos, T. M., Fernandes, V. H., Martínez-Capel, F., Muñoz-Mas, R., & Araújo, F. G. (2015). Application of the physical habitat simulation for fish species to assess environmental flows in an Atlantic Forest Stream in South-eastern Brazil. *Neotropical Ichthyology, 13*(4), 685–698. https://doi.org/10.1590/1982-0224-20140170

Da Rosa, V. C., & Garcia, M. (2015). Guildas ecológicas de diatomáceas (Bacillariophyta) epífitas em Acrostichum danaeifolium Langst. & Fisch em uma área úmida subtropical do sul do Brasil. *Acta Limnologica Brasiliensia, 27*(3), 311–321. https://doi.org/10.1590/S2179-975X4214

García De Jalón, D. (2008). *La Regulación de los caudales y su efecto en la biodiversidad*.

Garcia, T. D., Ota, R. R., Ferreira, D. G., Nascimento, R. H. C., Galindo, B. A., Pereira, L. S., & Zanatta, A. S. (2020). Distribution of siluriformes in a river under the influence of a small hydroelectric power plant of the Paraná river Basin, Brazil. *Iheringia - Serie Zoologia, 110*. https://doi.org/10.1590/1678-4766e2020005

Gleeson, T., & Richter, B. (2018). How much groundwater can we pump and protect environmental flows through time? Presumptive standards for conjunctive management of aquifers and rivers. *River Research and Applications, 34*(1), 83–92. https://doi.org/10.1002/rra.3185

Guerra, S., Lizárraga, L., & Návar, J. (2016). Riparian communities along longitudinal gradients in Mexico's Northeastern San Juan River. *Tecnología y Ciencias Del Agua, 7*(4), 42–79.

Haas, N. A., O'Connor, B. L., Hayse, J. W., Bevelhimer, M. S., & Endreny, T. A. (2014). Analysis of daily peaking and Run-Of-River operations with flow variability metrics, considering subdaily to seasonal time scales. *JAWRA Journal of the American Water Resources Association, 50*(6), 1622–1640. https://doi.org/10.1111/jawr.12228

Han, J.-C., Huang, G.-H., Zhang, H., Zhuge, Y.-S., & He, L. (2012). Fuzzy constrained optimization of eco-friendly reservoir operation using self-adaptive genetic algorithm: A case study of a cascade reservoir system in the Yalong River, China. *Ecohydrology, 5*(6), 768–778. https://doi.org/10.1002/eco.267

Katopodis, C. (2005). Developing a toolkit for fish passage, ecological flow management and fish habitat works. *Journal of Hydraulic Research, 43*(5), 451–467. https://doi.org/10.1080/00221680509500144

Kuriqi, A., Pinheiro, A. N., Sordo-Ward, A., & Garrote, L. (2020). Water-energy-ecosystem nexus: Balancing competing interests at a run-of-river hydropower plant coupling a hydrologic–ecohydraulic approach. *Energy Conversion and Management, 223*, 113267. https://doi.org/10.1016/j.enconman.2020.113267

Lee, S., Kim, J., & Hur, J. W. (2013). Assessment of ecological flow rate by flow duration and environmental management class in the Geum River, Korea. *Environmental Earth Sciences, 68*(4), 1107–1118. https://doi.org/10.1007/s12665-012-1812-y

Leite, M. G. P., Pimenta, E. C., Fujaco, M. A. G., & Eskinazi-Sant'Anna, E. M. (2016). Canais de irrigação na bacia do ribeirão do Melo (Municípios de Rio Espera e Capela Nova, Minas Gerais, Brasil): Habitats para a proliferação de Biomphalaria (Gastropoda: Planorbidae) e potencial disseminação da esquistossomose. *Brazilian Journal of Biology, 76*(3), 638–644. https://doi.org/10.1590/1519-6984.22714

Li, J., Qin, H., Pei, S., Yao, L., Wen, W., Yi, L., Zhou, J., & Tang, L. (2019). Analysis of an ecological flow regime during the Ctenopharyngodon Idella spawning period based on reservoir operations. *Water, 11*(10), 2034. https://doi.org/10.3390/w11102034

Liu, C., Zhao, C., Xia, J., Sun, C., Wang, R., & Liu, T. (2011). An instream ecological flow method for data-scarce regulated rivers. *Journal of Hydrology, 398*(1–2), 17–25. https://doi.org/10.1016/j.jhydrol.2010.11.026

Ma, L., Zhang, X., Wang, H., & Qi, C. (2018). Characteristics and practices of ecological flow in Rivers with flow reductions due to water storage and hydropower projects in China. *Water, 10*(8), 1091. https://doi.org/10.3390/w10081091

Ma, L., Wang, H., Qi, C., Zhang, X., & Zhang, H. (2019). Characteristics and adaptability assessment of commonly used ecological flow methods in water storage and hydropower projects, the case of Chinese River basins. *Water, 11*(10), 2035. https://doi.org/10.3390/w11102035

Magalhães Junior, A. P., & Curiel, P. B. (2017). Instrumentos de concesión administrativa de derechos de uso del agua en España. *Revista Brasileira de Recursos Hidricos, 22*. https://doi.org/10.1590/2318-0331.0217170011

Mantel, S., Hughes, D., & Muller, N. (2010). Ecological impacts of small dams on South African rivers Part 1: Drivers of change - water quantity and quality. *Water SA, 36*(3), 78–96.

Martínez, A., Manzanilla, S., & Zavalá, J. (2011). Vulnerability to climate change of marine and coastal fisheries in México. *Atmósfera, 24*(1), 103–123.

McKenna, J. E. (2019). The Laurentian Great Lakes: A case study in ecological disturbance and climate change. *Fisheries Management and Ecology, 26*(6), 486–499. https://doi.org/10.1111/fme.12317

Mwedzi, T., Katiyo, L., Mugabe, F. T., Bere, T., Bangira, C., Mangadze, T., & Kupika, O. L. (2016). A spatial assessment of stream-flow characteristics and hydrologic alterations, post dam construction in the Manyame catchment, Zimbabwe. *Water SA, 42*(2), 194–202. https://doi.org/10.4314/wsa.v42i2.03

Oliveira, V. H., Sousa, M. C., Morgado, F., & Dias, J. M. (2019). Modeling the impact of Extreme River discharge on the nutrient dynamics and dissolved oxygen in two adjacent estuaries (Portugal). *Journal of Marine Science and Engineering, 7*(11), 412. https://doi.org/10.3390/jmse7110412

Pal, S., & Sarda, R. (2021). Measuring the degree of hydrological variability of riparian wetland using hydrological attributes integration (HAI) histogram comparison approach (HCA) and range of variability approach (RVA). *Ecological Indicators, 120*, 106966. https://doi.org/10.1016/j.ecolind.2020.106966

Pal, S., Talukdar, S., & Ghosh, R. (2020). Damming effect on habitat quality of riparian corridor. *Ecological Indicators, 114*, 106300. https://doi.org/10.1016/j.ecolind.2020.106300

Pantoja, N. (2017). *Estimación de caudal ecológico mediante método hidrológicos, hidráulicos y ecológicos en la quebrada El Conejo (Mocoa-Putumayo).* Pontificia Universidad Javerian.

Patterson, L., Phelan, J., Goudreau, C., & Dykes, R. (2017). Flow-biology relationships based on fish habitat guilds in North Carolina. *Journal of the American Water Resources Association, 53*(1), 56–66. https://doi.org/10.1111/1752-1688.12495

Pedone-Bonfim, M. V. L., Alves da Silva, D., Maia, L. C., & Yano-Melo, A. M. (2018). Mycorrhizal benefits on native plants of the Caatinga, a Brazilian dry tropical forest. *Symbiosis, 74*, 79–88. https://doi.org/10.1007/s13199-017-0510-7

Perni, Á., Martínez-Paz, J., & Martínez-Carrasco, F. (2012). Social preferences and economic valuation for water quality and river restoration: The Segura River, Spain. *Water and Environment Journal, 26*(2), 274–284. https://doi.org/10.1111/j.1747-6593.2011.00286.x

Popa, F., Dumitran, G. E., Vuta, L. I., Tica, E. I., Popa, B., & Neagoe, A. (2020). Impact of the ecological flow of some small hydropower plants on their energy production in Romania. *Journal of Physics: Conference Series, 1426*(1), 12043. https://doi.org/10.1088/1742-6596/1426/1/012043

Qin, L., Li, K., Li, Y., Liang, R., Chen, M., & Hodges, B. R. (2016). A habitat similarity model based on vague sets to assess Schizothorax prenanti spawning habitat. *Ecological Engineering, 96*, 86–93. https://doi.org/10.1016/j.ecoleng.2016.04.007

Roets, W., Raitt, L., & Brendonck, L. (2008). Groundwater discharges to aquatic ecosystems associated with the Table Mountain Group (TMG) aquifer: A conceptual model. *Water SA, 34*(1), 12–45.

Rosa, B. F. J. V., Martins, R. T., & Alves, R. G. (2015). Distribuição de oligoquetas em um riacho da mata atlântica, sudeste do Brasil. *Brazilian Journal of Biology, 75*(1), 1–7. https://doi.org/10.1590/1519-6984.02313

Talukdar, S., Pal, S., Chakraborty, A., & Mahato, S. (2020). Damming effects on trophic and habitat state of riparian wetlands and their spatial relationship. *Ecological Indicators, 118*, 106757. https://doi.org/10.1016/j.ecolind.2020.106757

Tan, Y.-Y., Wang, X., Li, C.-H., Cai, Y.-P., Yang, Z.-F., & Wang, Y.-L. (2012). Estimation of ecological flow requirement in Zoige Alpine Wetland of southwest China. *Environmental Earth Sciences, 66*(5), 1525–1533. https://doi.org/10.1007/s12665-011-1392-2

Theodoropoulos, C., Papadaki, C., Vardakas, L., Dimitriou, E., Kalogianni, E., & Skoulikidis, N. (2019). Conceptualization and pilot application of a model-based environmental flow assessment adapted for intermittent rivers. *Aquatic Sciences, 81*(1), 10. https://doi.org/10.1007/s00027-018-0605-0

Thürer, M., Tomašević, I., Stevenson, M., Qu, T., & Huisingh, D. (2018). A systematic review of the literature on integrating sustainability into engineering curricula. *Journal of Cleaner Production, 181*, 608–617. https://doi.org/10.1016/j.jclepro.2017.12.130

Tibúrcio, G. S., Da Silva Carvalho, C., Ferreira, F. C., Goitein, R., & Ribeiro, M. C. (2016). Efeito da paisagem na ocorrência da ictiofauna em rios de primeira ordem da região sudeste do Brasil. *Acta Limnologica Brasiliensia, 28*. https://doi.org/10.1590/S2179-975X2515

Tsheboeng, G. (2020). Environmental factors that influence species diversity of floodplain plant communities in different flooding phases in the Okavango Delta, Botswana. *African Biodiversity & Conservation, 50*(1), 120–139.

Upadhyay, R., Dasgupta, N., Hasan, A., & Upadhyay, S. K. (2011). Managing water quality of river Yamuna in NCR Delhi. *Physics and Chemistry of the Earth, 36*(9–11), 372–378. https://doi.org/10.1016/j.pce.2010.03.018

Walsh, G., & Wepener, V. (2009). The influence of land use on water quality and diatom community structures in urban and agriculturally stressed rivers. *Water SA, 35*(5), 579–594. https://doi.org/10.4314/wsa.v35i5.49184

Wang, Y., Zhang, N., Wang, D., Wu, J., & Zhang, X. (2018). Investigating the impacts of cascade hydropower development on the natural flow regime in the Yangtze River, China. *Science of the Total Environment, 624*, 1187–1194. https://doi.org/10.1016/j.scitotenv.2017.12.212

Wang, L., Chen, Q., Zhang, J., Xia, J., Mo, K., & Wang, J. (2020). Incorporating fish habitat requirements of the complete life cycle into ecological flow regime estimation of rivers. *Ecohydrology, 13*(4), e2204. https://doi.org/10.1002/eco.2204

Weerts, S. P., & Cyrus, D. P. (2008). Inter-basin water transfer effects on the invertebrates of the Mvuzane River: A test of the SASS rapid biomonitoring protocol to assess the impact of flow modification. *Water SA, 34*(2), 175–182. https://doi.org/10.4314/wsa.v34i2.183637

Westwood, C. G., England, J., Dunbar, M. J., Holmes, N. T. H., Leeming, D. J., & Hammond, D. (2017). An approach to setting ecological flow thresholds for southern English chalk streams. *Water and Environment Journal, 31*(4), 528–536. https://doi.org/10.1111/wej.12275

Xu, G., Sun, H. L., & Li, S. C. (2014). Research on the ecological scheduling decisions of the Jinping cascade hydropower station on the Yalong river. *Energy Sources, Part A: Recovery, Utilization and Environmental Effects, 36*(10), 1115–1122. https://doi.org/10.1080/15567036.2010.544010

Yan, Z., Zhou, Z., Sang, X., & Wang, H. (2018). Water replenishment for ecological flow with an improved water resources allocation model. *Science of the Total Environment, 643*, 1152–1165. https://doi.org/10.1016/j.scitotenv.2018.06.085

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